Loglinear Models for Counts

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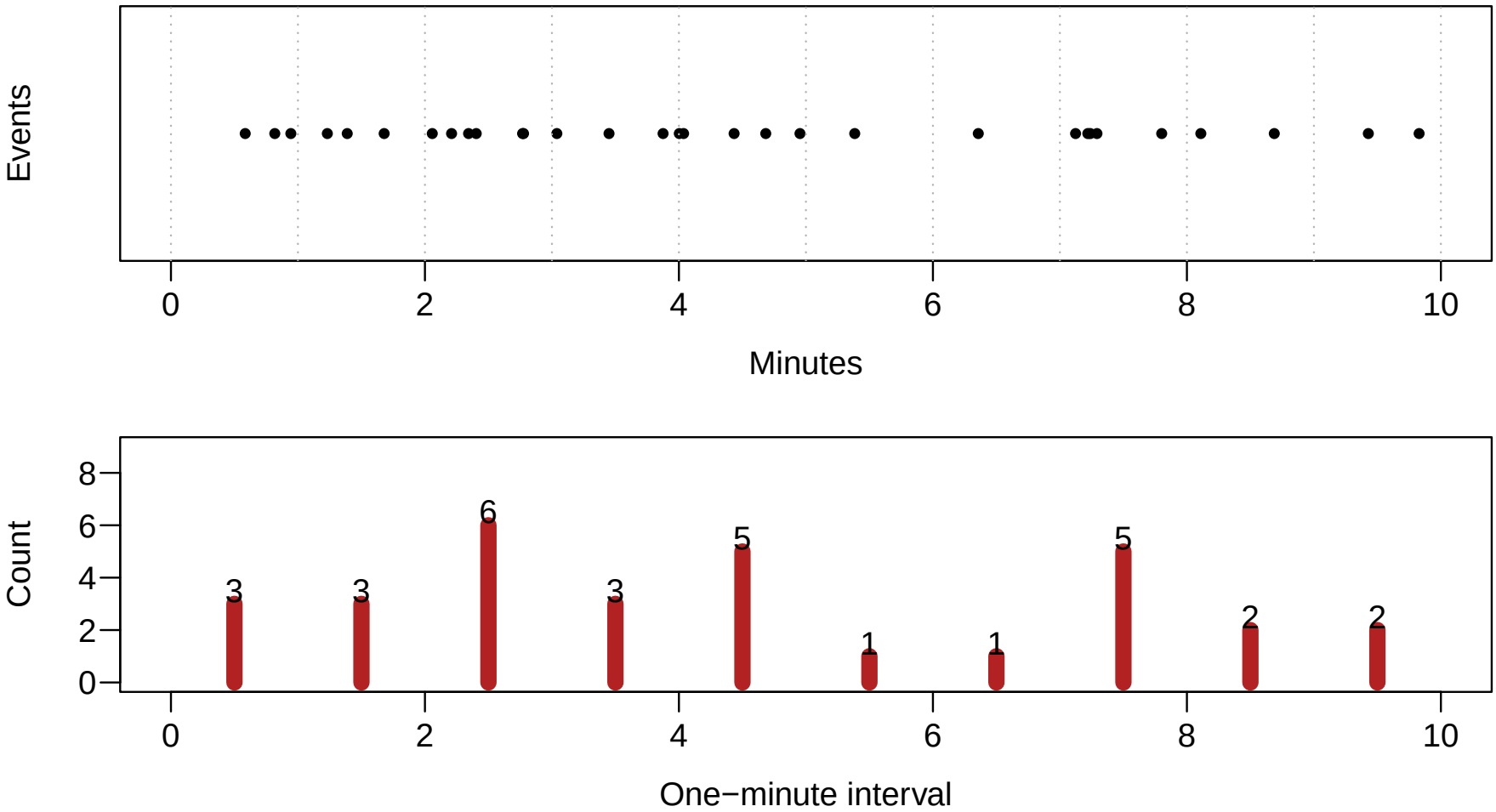
Counting Events

Suppose we count arrivals during a one-minute interval.

- What values can the response take?
- What would it mean for the average count to be 2.5?
- If the underlying rate stays the same, should every minute have the same count?

What would a probability model for this response need to describe?

Events and Counts over Time



What if we changed our focus to two-minute intervals?

The Poisson Distribution

For $Y \sim \text{Poisson}(\mu)$,

$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

- Y is a count, with no fixed upper bound.
- $\mu > 0$ is the expected count for the exposure being observed.
- μ need not be an integer.
- Assume the events are independent and cannot occur simultaneously

Poisson Probabilities

$Y \sim \text{Poisson}(\mu)$,

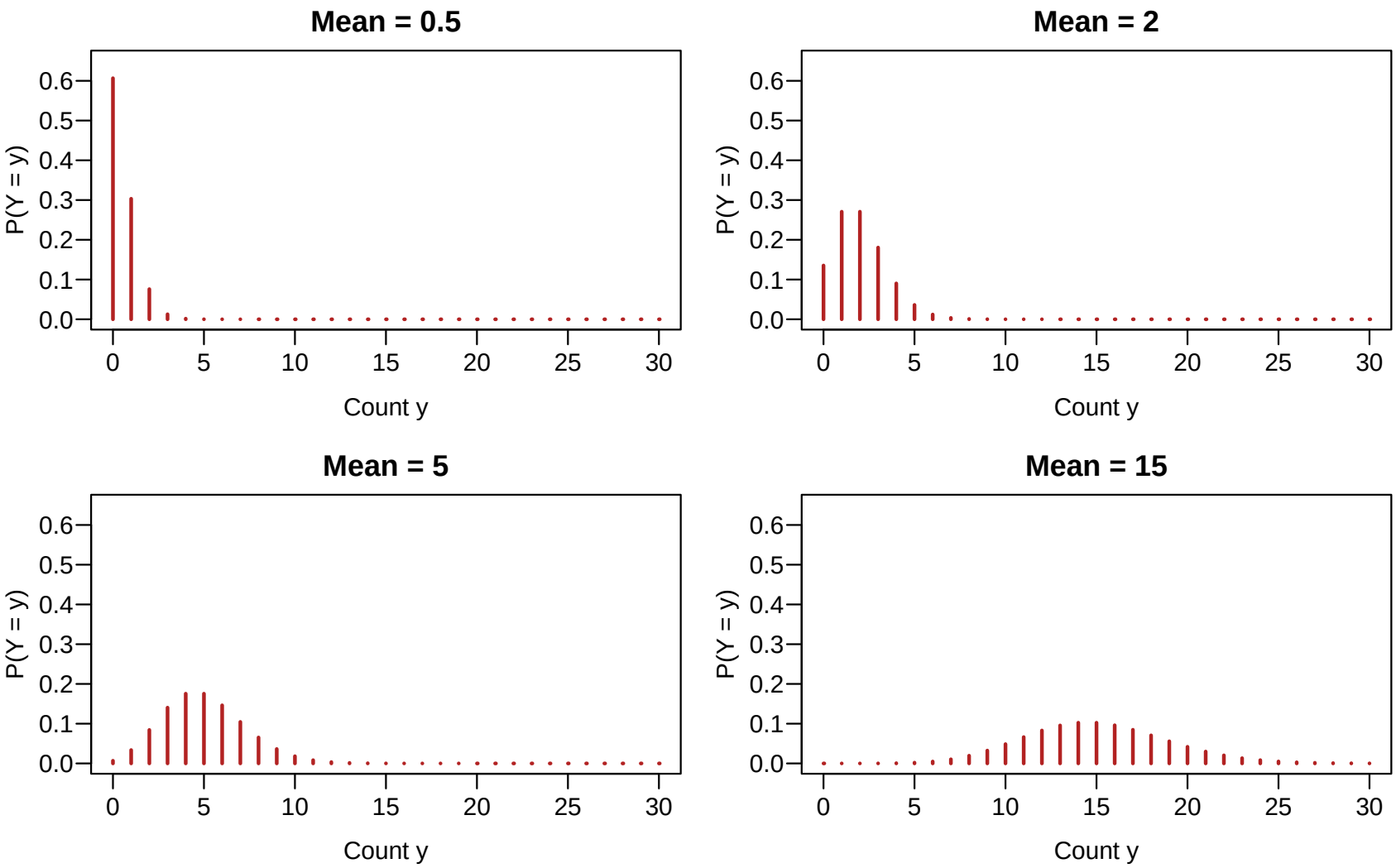
$$P(Y = y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

Because the Poisson is a discrete distribution, each non-zero count has a positive probability.

For $\mu = 2$, find:

- $P(Y = 0)$
- $(1 < Y < 3)$
- $P(Y > 4)$

Changing the Expected Count



As μ increases, the Poisson becomes approximately normal

A Connection to the Binomial

Compare binomial counts with the same mean:

$$\text{Bin}(10, 0.2), \quad \text{Bin}(100, 0.02), \quad \text{Bin}(1000, 0.002).$$

- Many opportunities, each with a small event probability.
- With $np = \mu$ fixed, the binomial approaches Poisson(μ) as n grows and p shrinks.
- The binomial variance $np(1-p)$ approaches μ.

What distinguishes a binomial count from a Poisson count?

Mean and Variance

$$E(Y) = \mu, \qquad \text{Var}(Y) = \mu, \qquad \text{SD}(Y) = \sqrt{\mu}.$$

Like the binomial, the mean and variance share a distributional parameter.

This can have impact on Wald statistic:

- Case when $Y = 0$
- CI can contain negative counts
- Only boundary is near zero, unlikely with larger counts

Exposure and the Mean Count

At a constant rate λ per minute, the expected count in t minutes is

$$\mu = \lambda t.$$

- Doubling exposure doubles the expected count.
- For independent Poisson counts, adding intervals adds their means.
- A count of 8 in two minutes and a count of 8 in ten minutes suggest different rates.

When Would Poisson Be Reasonable?

- Arrivals in equal intervals at a stable rate
- Students arriving at the student union
- Number of floods in a ten year period
- Number of earthquakes in a ten year period

Which situations might have more variability than a simple Poisson model allows? Why?

Modeling Counts in a Contingency Table

Previously, we modeled an individual's response probability.

Now each **cell count** is a response:

$$n_{ij} \sim \text{Poisson}(\mu_{ij}).$$

- X and Y identify the row and column categories.
- n_{ij} is the observed count in cell (i, j).
- μ_{ij} is that cell's expected count.

The classification variables are treated symmetrically. Neither must be designated as the outcome variable.

Independence and Expected Counts

Under independence,

$$\pi_{ij} = \pi_{i+}\pi_{+j}.$$

For a table with total n,

$$\mu_{ij} = n\pi_{ij} = n\pi_{i+}\pi_{+j}.$$

Taking logs gives an additive relationship:

$$\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}.$$

Which parts depend on the row? The column?

The Loglinear Independence Model

We write the model as

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y$$

- Poisson random component and log link.
- Row and column effects add on the log-mean scale.
- Effects multiply on the expected-count scale.
- No row-by-column interaction: independence of X and Y.

The superscripts X and Y label variables; they are not powers.

Happiness and Belief in Heaven

General Social Survey respondents reported happiness and whether they believed in heaven.

	Belief in Heaven	
Happiness	No	Yes
Not too happy	32	190
Pretty happy	113	611
Very happy	51	326

Three happiness categories and two belief categories give six cell counts to model.

The Independence Model in R

```
> fit <- glm(count ~ happy + heaven, family = poisson,
+            data = HappyHeaven)
> summary(fit)
Call:
glm(formula = count ~ happy + heaven, family = poisson, data =
    HappyHeaven)
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)    3.4931     0.0941   37.13  < 2e-16 ***
happypretty    1.1821     0.0767   15.41  < 2e-16 ***
happyvery      0.5296     0.0846    6.26  3.9e-10 ***
heavenyes      1.7492     0.0774   22.60  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for poisson family taken to be 1)
    Null deviance: 1019.87238  on 5  degrees of freedom
Residual deviance:    0.89111  on 2  degrees of freedom
AIC: 49.5
Number of Fisher Scoring iterations: 3
```

Observed and Fitted Counts

```
> HappyHeaven$fitted <- fitted(fit)
> HappyHeaven
   happy heaven count  fitted
1    not     no    32  32.889
2    not    yes   190 189.111
3 pretty     no   113 107.259
4 pretty    yes   611 616.741
5   very     no    51  55.852
6   very    yes   326 321.148
```

Do the fitted counts look familiar from chi-squared tests of independence?

χ^2 Test of Independence

```
> tab
        heaven
happy      no yes
  not      32 190
  pretty  113 611
  very     51 326

> chisq.test(tab)

  Pearson's Chi-squared test

X-squared = 0.884, df = 2, p-value = 0.64

> ## Deviance from model
> c(dev = deviance(fit), df = df.residual(fit))
     dev       df
0.89111 2.00000
```

Coefficients as Odds

For the full model, we have too many parameters

$$\log(\mu_i) = \lambda + \lambda_1^X + \lambda_2^X$$

```
> df <- data.frame(Drug = c("A", "B"), y = c(12, 24))
> glm(y ~ Drug, family = poisson, data = df)

Call:  glm(formula = y ~ Drug, family = poisson, data = df)

Coefficients:
(Intercept)        DrugB
      2.485        0.693

Degrees of Freedom: 1 Total (i.e. Null);  0 Residual
Null Deviance:      4.08
Residual Deviance: -0.0000000000000355   AIC: 13.4
```

What is the interpretation of λ_2^X once λ_1^X is the reference?

Interpreting the Coefficients

(Intercept)	happypretty	happyvery	heavenyes
3.49	1.18	0.53	1.75
32.89	3.26	1.70	5.75

The fitted model has reference categories `not` (not too happy) and `no` belief in heaven.

- $e^{3.4931} \approx 32.89$: expected count in the reference cell.
- $e^{1.1821} \approx 3.26$: pretty/not expected-count ratio within either heaven category.
- $e^{1.7492} \approx 5.75$: yes/no expected-count ratio within each happiness category.

The last ratio is the odds of belief in heaven in each row. The model assumes those odds are the same across happiness levels.

Association as an Interaction

Add row-by-column terms:

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}.$$

- The column effect may now depend on the row.
- There are $(r-1)(c-1)$ additional parameters.
- Total parameters: $1 + (r-1) + (c-1) + (r-1)(c-1) = rc$.

This is the saturated model: one fitted mean per cell.

What is the interpretation of λ_{ij}^{XY} in this case?

The Saturated Model in R

```
> fit_sat <- glm(count ~ happy * heaven, family = poisson,
+                data = HappyHeaven)
> summary(fit_sat)
Call:
glm(formula = count ~ happy * heaven, family = poisson, data =
    HappyHeaven)
Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)             3.4657     0.1768   19.61   <2e-16 ***
happypretty             1.2617     0.2002    6.30    3e-10 ***
happyvery               0.4661     0.2255    2.07    0.039 *
heavenyes               1.7813     0.1911    9.32   <2e-16 ***
happypretty:heavenyes  -0.0936     0.2168   -0.43    0.666
happyvery:heavenyes     0.0738     0.2433    0.30    0.762
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Dispersion parameter for poisson family taken to be 1)
    Null deviance: 1.0199e+03  on 5  degrees of freedom
Residual deviance: 2.2204e-16  on 0  degrees of freedom
AIC: 52.61
Number of Fisher Scoring iterations: 2
```

Using Deviance to Test Independence

Just as before, we can also check deviance to test against the interaction term

```
> anova(fit, fit_sat, test = "LRT")
Analysis of Deviance Table

Model 1: count ~ happy + heaven
Model 2: count ~ happy * heaven
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         2      0.891
2         0      0.000  2    0.891     0.64
```

Review: Two-Way Tables

1. What is the response in a loglinear model?
2. Why does independence imply an additive log-mean model?
3. Why can a main effect be highly significant even when the variables are independent?
4. What restriction disappears when we add an interaction?